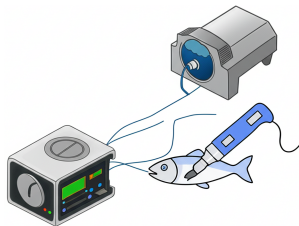


Deep Learning for Marine Biomass Analysis

Using Rapid Evaporative Ionization Mass Spectrometry



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Fish $\approx$ Marine Biomass

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Figure 1. Hoki (Blue Grenadier)



Figure 2. Mackerel

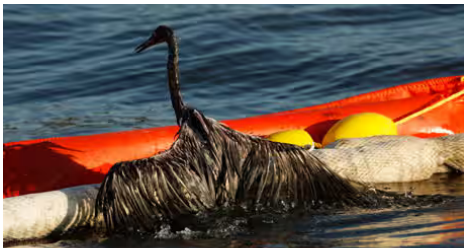
The Problem: Challenges in the Global Seafood Industry

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Nutrition Facts	
6 servings per container	
Serving size	4-5 ounces(187g)
Amount per serving	
Calories	200
% Daily Value*	
Total Fat 5g	6%
Saturated Fat 0.5g	3%
Trans Fat 0g	
Cholesterol 80mg	27%
Sodium 610mg	27%
Total Carbohydrate 10g	4%
Dietary Fiber 0g	0%
Total Sugars 5g	
Includes 0g Added Sugars	0%
Protein 27g	
Vitamin D 2mcg	10%
Calcium 78mg	6%
Iron 3mg	15%
Potassium 518mg	10%
*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.	



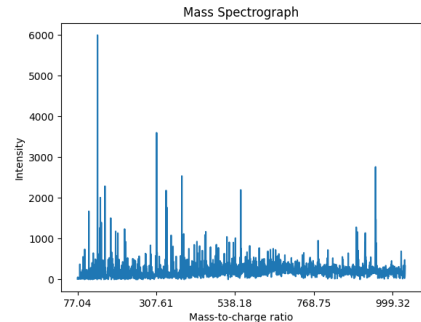
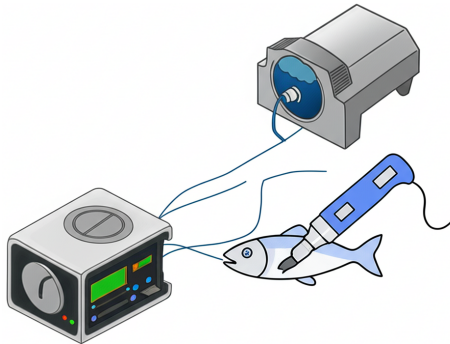
How can we rapidly and accurately identify the state of a fish sample in a processing plant?

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Traditional lab methods are slow and expensive. We need an instant, on-site solution. Rapid Evaporative Ionization Mass Spectrometry (REIMS) provides a chemical 'fingerprint', but the data is high-dimensional, noisy, and complex.



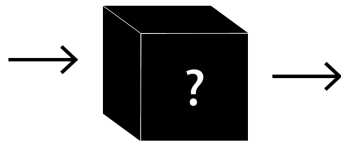
My Solution - Building a Better "Fingerprint" Reader

Traditional ML struggles. My solution: Use Deep Learning to decode these complex chemical fingerprints.

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I developed a suite of novel methods based on Transformers and Self-Supervised Learning to tackle these challenges head-on.

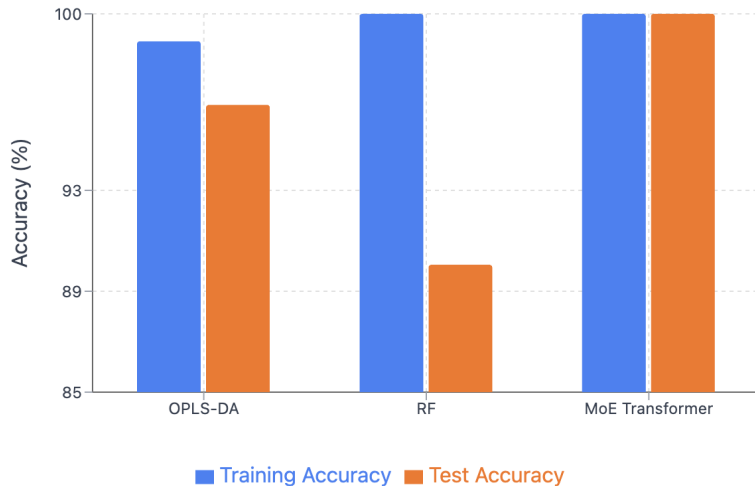
First, I taught the model to identify fish species with 100% accuracy.

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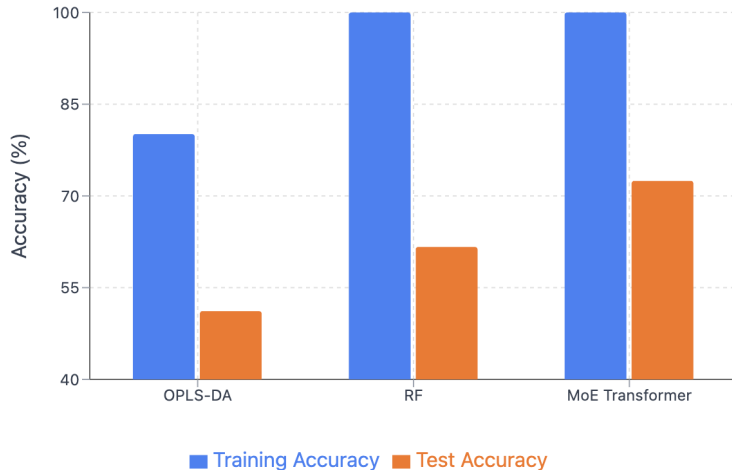
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Fish Species Classification Accuracy Comparison



Then, I taught the model to identify fish body parts with 72% accuracy.

Fish Body Parts Classification Accuracy Comparison

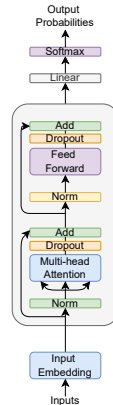


I used a Transformer architecture, famous for its use in language models.

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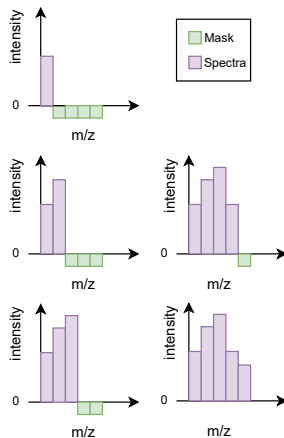
I treated the mass spectra data like a sentence. The Transformer is exceptionally good at finding the relationships between different parts of the sequence (the 'words' of the chemical fingerprint) to understand its meaning.

To improve the transformer performance, I developed a novel pre-training strategy

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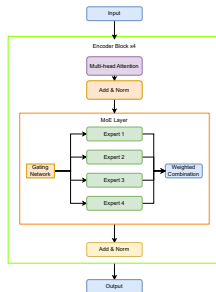


To handle diverse data, I designed a Mixture of Experts (MoE) Transformer

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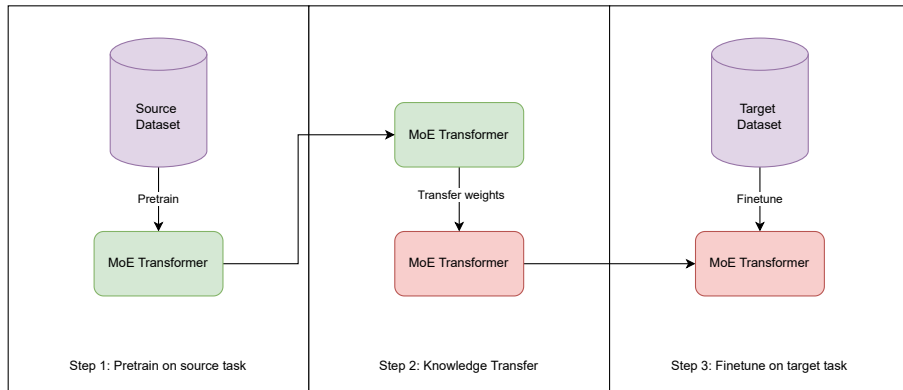
Instead of one giant model, I trained multiple 'specialist' models. A 'gating network' learns to send a specific fish sample to the right expert for analysis. This is what I called the 'Gone Phishing' model.

To handle the lack of data, I applied Transfer Learning to the MoE Transformer

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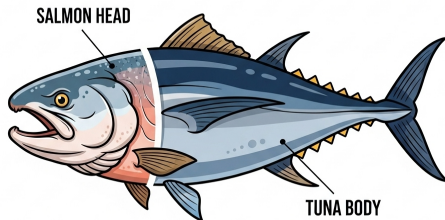
Transfer learning trains a network on a source task, then transfers those weights to a new network, and fine-tunes on a target task. Essentially transferring any shared knowledge between tasks.

Then, I tackled the difficult task of detecting cross-species adulteration

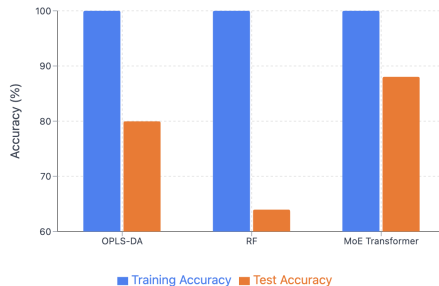
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Cross-species Adulteration Accuracy Comparison

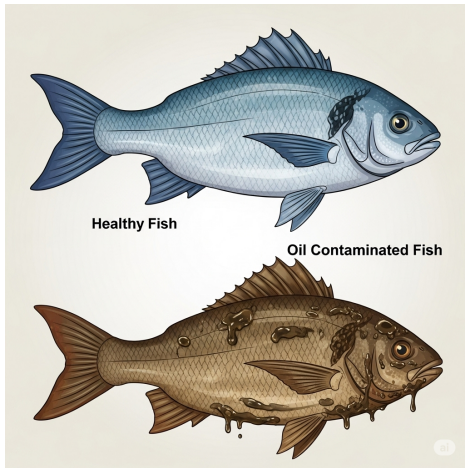


Next, I tackled the difficult task of detecting oil contamination

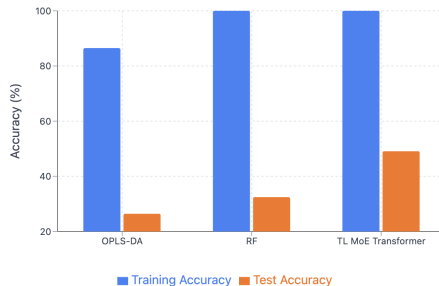
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Oil Contamination Detection Accuracy Comparison



The final challenge was solving an unaddressed problem: identifying a fish's processing batch.

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NEW ZEALAND / FOOD

Frozen anchovies in recall over biotoxin threat

6:23 pm on 25 April 2025

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This is critical for food safety and recalls, but no existing method could do it.

I developed 'SpectroSim,' a self-supervised model to solve this without labels

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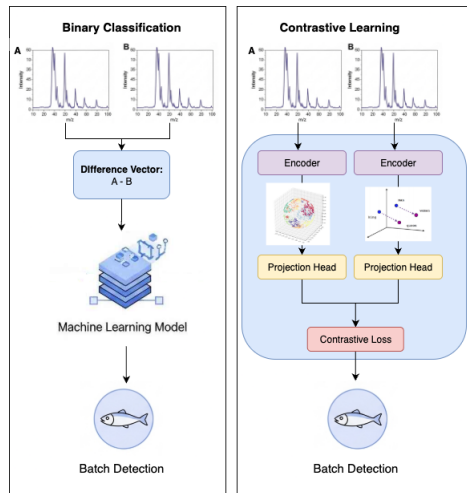


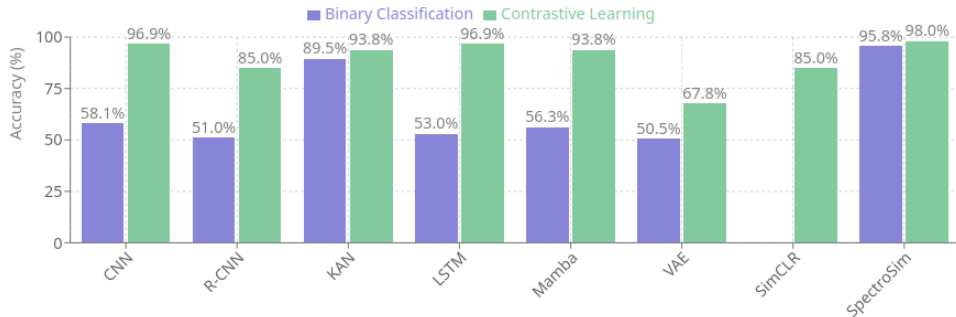
Figure 5. Caption

SpectroSim achieved 98% accuracy, dramatically outperforming all other approaches.

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This demonstrates that for unlabeled, complex matching tasks, self-supervised contrastive learning is the superior strategy.

Summary of Contributions and Technical Skills

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- ◇ **Neural Network:** Developed novel Deep Learning architectures (Transformers, MoEs) for complex, noisy 1D signal data.
- ◇ **Data Transfer:** Successfully applied Transfer Learning and Self-Supervised learning (SpectroSim) to solve low-data and unlabeled problems.
- ◇ **Code:** Implemented end-to-end projects, from data analysis and preprocessing to model training and validation.
- ◇ **Impact:** Created solutions with direct real-world applications in industrial automation, quality control, and food safety.

Conclusion & Impact

Thanks for your attention!

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